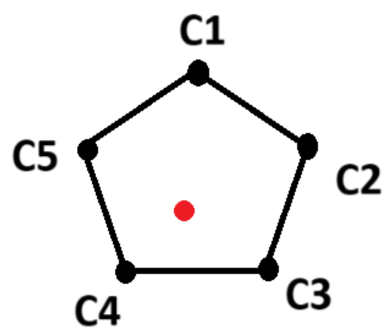
Brac University
Department of Computer Science and Engineering
CSE423: Computer Graphics
Assignment 04

1.Write three comparisons between ambient, diffuse, and specular reflection. [3]

2. Point P(6, 3, 4) lies on a street crosswalk with ambient light intensity 4. The reflection coefficients are: Ambient: 0.3, Diffuse: 0.6, Specular: 0.15. A street lamp at L(1, 1, 25) has intensity 12.

   a. Given the radius of influence = 30 units, find out the Attenuation. [4]
   b. Calculate total illumination at P on the yz plane using Phong's model. Viewer is at V(8, 8, 8), shininess factor 50. [8]



3. Given that, C1= 12, C2= (C1*2), C3=(C4+10), C4 =(C2-5), C5=(C3*2)
Which shading method is most suitable for finding the color of the red pixel in the pentagon? Briefly explain the principle and provide the mathematical expressions for the pixel's color.[5]